

M/C	Written	Total
10 / 11	18.5 / 19	28.5 / 30

KNOWLEDGE

True/False

Indicate whether the statement is true or false. (0.5 mark each)

F

1. A solution to the trigonometric equation
- $\sin x + \cos x = 0$
- is
- $x = 0$
- .

F

2. The function
- $y = -3 \sin x + 1$
- has an amplitude of
- -3
- .

F

3. The graph of
- $y = \sin\left(\frac{1}{2}x + \frac{\pi}{4}\right)$
- can be drawn by translating the graph of
- $y = \sin x$
- left
- $\frac{\pi}{4}$
- radians.

(2) +

4. The graph of
- $y = a \sin x$
- that passes through the point
- $\left(\frac{\pi}{2}, -2\right)$
- has an amplitude of 2.

Application

1. A pebble is embedded in the tread of a rotating bicycle wheel of diameter 60 cm. If the wheel rotates at 4 revolutions per second, determine a relationship between the height,
- h
- , of the pebble above the ground, in centimetres, as a function of time
- t
- , in seconds.

$$D = 60 \text{ cm}$$

$$4 \text{ rev/s} = \frac{1}{4} \text{ s/rev} = p$$

(2)

$$r = 30 \text{ cm} = a = y$$

$$k = \frac{2\pi}{\frac{1}{4}} = 8\pi$$

$$h(t) = -30 \cos(8\pi t) + 30$$

2. Sketch the function,
- $y = -\sec\left(\frac{1}{3}x - \frac{\pi}{2}\right) - 2$
- , for one cycle.

Label axis properly. Use mapping rule.

$$y = -\sec\left[\frac{1}{3}\left(x - \frac{3\pi}{2}\right)\right] - 2$$

(4)

$$(x, y) \rightarrow \left(3x + \frac{3\pi}{2}, -y - 2\right)$$

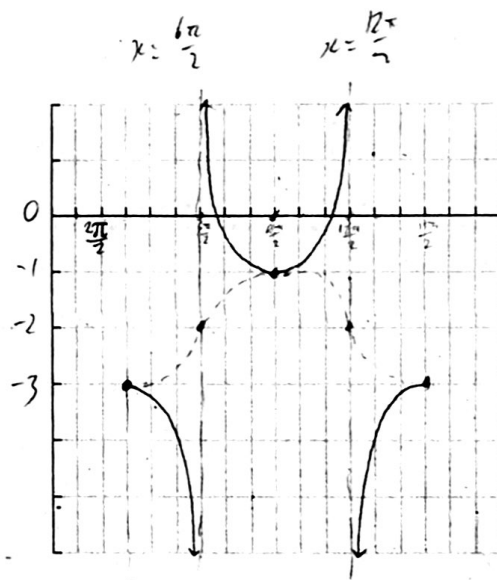
$$(0, 1) \rightarrow \left(\frac{3\pi}{2}, -3\right)$$

$$\left(\frac{\pi}{2}, 0\right) \rightarrow \left(\frac{6\pi}{2}, -2\right)$$

$$(\pi, -1) \rightarrow \left(\frac{9\pi}{2}, -1\right)$$

$$\left(\frac{3\pi}{2}, 0\right) \rightarrow \left(\frac{12\pi}{2}, -2\right)$$

$$(2\pi, 1) \rightarrow \left(\frac{15\pi}{2}, -3\right)$$

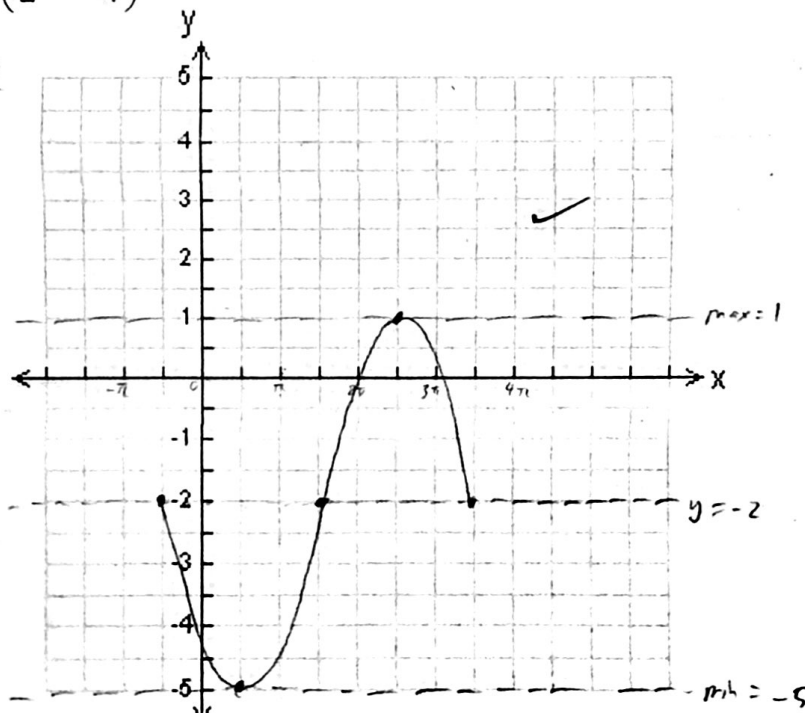


3. Sketch
- $f(x) = -3 \sin\left(\frac{1}{2}x + \frac{\pi}{4}\right) - 2$
- for one cycle using 5 point method.

$$f(x) = -3 \sin\left[\frac{1}{2}\left(x + \frac{\pi}{2}\right)\right] - 2$$

$$k = \frac{1}{2} \quad p = \frac{2\pi}{\frac{1}{2}} = 4\pi$$

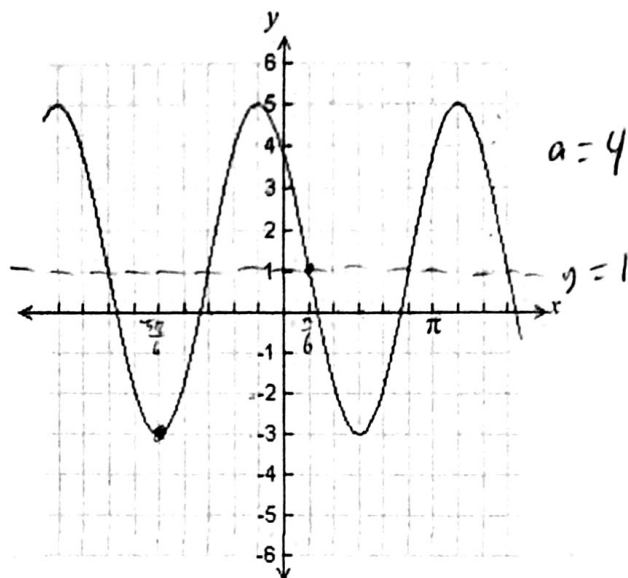
$$d = -\frac{\pi}{2}$$

(3)

4. Use the given graph to determine the following:

Sine equation: $y = -4 \sin\left[\frac{3}{2}\left(x - \frac{41\pi}{6}\right)\right] + 1$

Cosine equation: $y = -4 \cos\left[\frac{3}{2}\left(x + \frac{61\pi}{6}\right)\right] + 1$



(3) $p = \frac{2\pi}{6} = \frac{4\pi}{3}$ $\frac{\pi}{6} + 5\left(\frac{4\pi}{3}\right) = \frac{41\pi}{6}$

$k = \frac{2\pi}{4\pi/3} = \frac{3}{2}$ $-\frac{5\pi}{6} - 7\left(\frac{4\pi}{3}\right) = -\frac{61\pi}{6}$

5. Write the equation for the cosine function having an amplitude of 4, period of $\frac{7\pi}{3}$, reflected in the x-axis, phase shift of $\frac{\pi}{3}$ to the left, and vertical translation of 3 units downward.

$p = \frac{7\pi}{3}$

(2)

$k = \frac{2\pi}{7\pi/3} = \frac{6}{7}$

$y = -4 \cos\left[\frac{6}{7}\left(x + \frac{\pi}{3}\right)\right] - 3$

6. A square is cut by starting at one corner (A) and cutting to the other side somewhere along side B. Determine the range of values that angle x can be and determine the range of values that side y can be.

$\tan x_{\max} = \frac{B}{B}$

$y_{\max} = \sqrt{B^2 + B^2} = \sqrt{2}B$

(2.5)

$x_{\max} = \tan^{-1}(1)$

$= \sqrt{2}B$

(3)

$x_{\max} = \frac{\pi}{4}$

$= \sqrt{2}B$

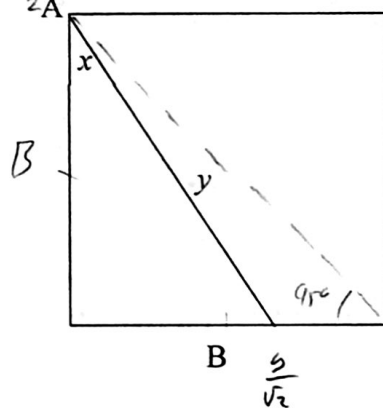
$\tan x_{\min} = \frac{0}{B}$

$y_{\min} = \sqrt{0^2 + B^2}$

$x_{\min} = \tan^{-1}(0)$

$= B$

$x_{\min} = 0$



Range x : $\{x \in \mathbb{R} \mid 0 \leq x \leq \frac{\pi}{4}\}$

Range y : $\{y \in \mathbb{R} \mid B \leq y \leq \sqrt{2}B\}$